

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is light green. Both are tilted at an angle.

# Command Line: Network and Link Layer Utilities



## Topics to be covered in this Tutorial:

- route
- arp
- nmap
- ssh
- scp



# route command

**route** command in Linux is used when you want to work with the IP/kernel routing table. It is mainly used to set up static routes to specific hosts or networks via an interface. It is used for showing or updating the IP/kernel routing table.

Basically if you want to show / manipulate the IP routing table.

```
$sudo apt-get install net-tools
```



## ***What is a Routing Table?***

A routing table is a file containing information on how the information or packets should be transferred: the network path to all nodes or devices within a network.

**It is a map between the destination IP address and the gateway/next-hop.** The hop-by-hop routing is widely used, the packet contains the routing table to reach the next hop, once reached, it will read the routing table again to reach the next hop.

Hence, by using the route command you can communicate with subnets and different networks, you can also block the traffic between networks or devices by modifying the routing table.

- n : displays routing table in full numeric form.
- v: displays the routing table with verbose information.

```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ route  
Kernel IP routing table  
Destination      Gateway          Genmask          Flags Metric Ref    Use Iface  
default          _gateway        0.0.0.0          UG    600    0      0 wlp13s0  
link-local      0.0.0.0         255.255.0.0      U     1000   0      0 wlp13s0  
192.168.0.0      0.0.0.0         255.255.255.0    U     600    0      0 wlp13s0  
ubuntu@ubuntu:~$ route -n  
Kernel IP routing table  
Destination      Gateway          Genmask          Flags Metric Ref    Use Iface  
0.0.0.0          192.168.0.1     0.0.0.0          UG    600    0      0 wlp13s0  
169.254.0.0      0.0.0.0         255.255.0.0      U     1000   0      0 wlp13s0  
192.168.0.0      0.0.0.0         255.255.255.0    U     600    0      0 wlp13s0  
ubuntu@ubuntu:~$
```



We can assign a default gateway using the following command:

```
ubuntu@ubuntu:~$ sudo route add default gw 169.254.0.0
ubuntu@ubuntu:~$ route -n
Kernel IP routing table
Destination      Gateway          Genmask          Flags Metric Ref    Use Iface
0.0.0.0          169.254.0.0     0.0.0.0          UG      0      0        0 wlp13s0
0.0.0.0          192.168.0.1     0.0.0.0          UG      600    0        0 wlp13s0
169.254.0.0      0.0.0.0         255.255.0.0      U        1000   0        0 wlp13s0
192.168.0.0      0.0.0.0         255.255.255.0    U        600    0        0 wlp13s0
ubuntu@ubuntu:~$
```

This assigns a gateway address on which all the packets that do not belong to the network are forwarded.



We can reject routing to a particular host or network, using the **reject** command.

```
ubuntu@ubuntu:~$ sudo route add -host 192.168.1.51 reject
ubuntu@ubuntu:~$ ping 192.168.1.51
ping: connect: No route to host
ubuntu@ubuntu:~$
```

Now if you will ping to the above-mentioned IP it will display “Network is unreachable”.



## arp command

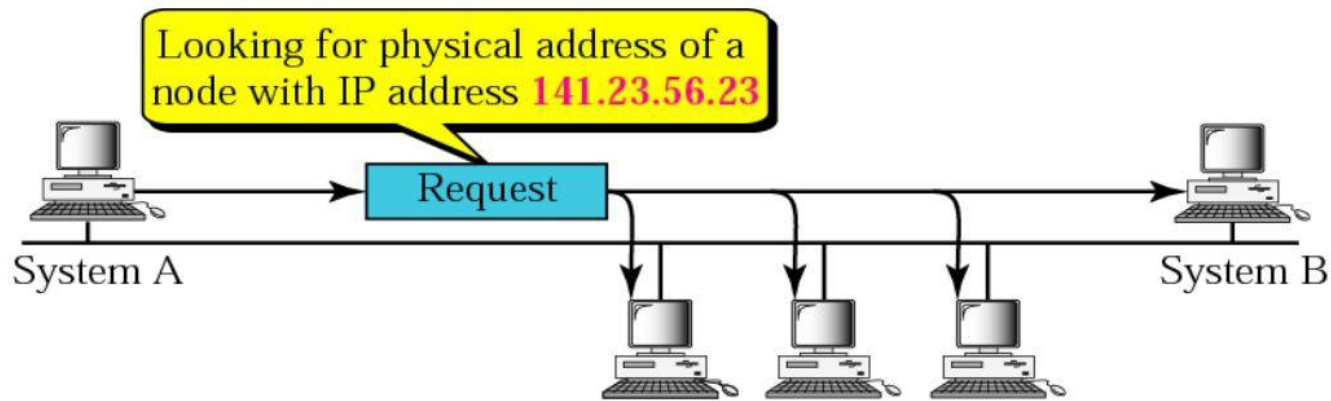
**arp command** manipulates the System's ARP cache. It also allows a complete dump of the ARP cache.

**ARP** stands for Address Resolution Protocol. The primary function of this protocol is to resolve the IP address of a system to its mac address, and hence it works between level 2(Data link layer) and level 3(Network layer).

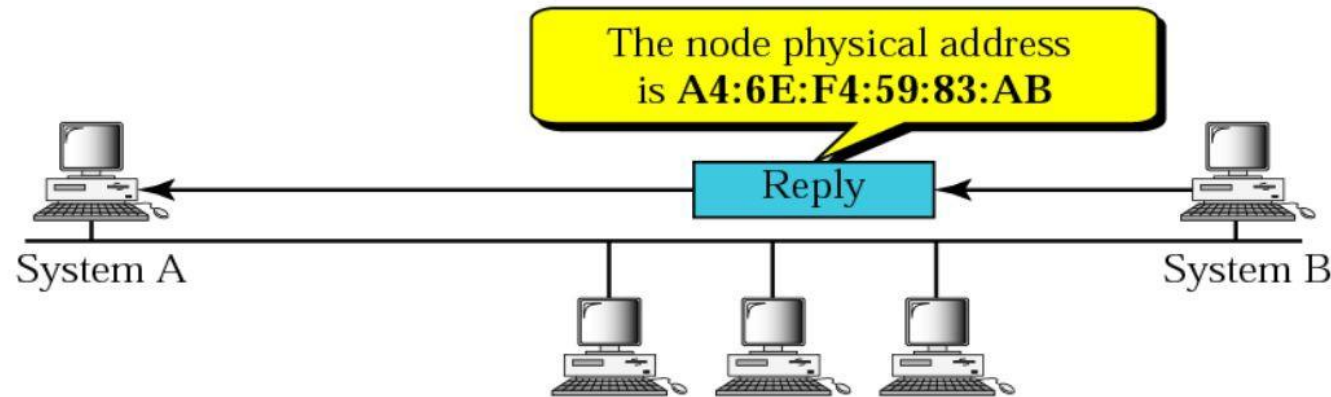


The Sender asks the receiver to announce its physical address when needed. ARP is designed to this purpose. Anytime, a host(router) needs to find the physical address of another host(router) on its network, it sends an ARP Request and the query packet is broadcast over the network.

Host will ignore the ARP request, if the IP address is not the host. Otherwise the host will reply to the sender. Every host(router) on the network receives and processes the ARP query packet. But only the intended recipient recognizes its IP address and sends back an ARP Reply.



a. ARP request is broadcast



b. ARP reply is unicast



The `arp` program displays and modifies the Internet-to-MAC address translation tables used by the address resolution protocol.

With no flags, the program displays the current ARP entry for hostname. The host may be specified by name or by number, using Internet dot notation.

Options that modify the ARP translation tables (`-d`, `-f`, and `-s`) can be used only when the invoke command is granted the `sys_net_config` privilege.

Source: [https://docs.oracle.com/cd/E88353\\_01/html/E72487/arp-8.html](https://docs.oracle.com/cd/E88353_01/html/E72487/arp-8.html)



**arp cache** is basically the memory location where we have recently resolved IP and mac addresses.

```
root@kali:~# arp -a
_gateway (10.0.2.1) at 52:54:00:12:35:00 [ether] on eth0
root@kali:~# ping 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=0.541 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=0.280 ms
^C
--- 10.0.2.4 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1031ms
rtt min/avg/max/mdev = 0.280/0.410/0.541/0.132 ms
root@kali:~# arp -a
? (10.0.2.4) at 08:00:27:ad:87:b3 [ether] on eth0
_gateway (10.0.2.1) at 52:54:00:12:35:00 [ether] on eth0
root@kali:~#
```

- **-v, --verbose:** This option shows the verbose information.
- **-n, --numeric:** This option shows numerical addresses instead of symbolic host, port or usernames

```
root@kali:~# arp -v
```

| Address  | HWtype | HWaddress         | Flags | Mask | Iface |
|----------|--------|-------------------|-------|------|-------|
| machine1 | ether  | 08:00:27:ad:87:b3 | C     |      | eth0  |
| machine2 | ether  | 08:00:27:27:d6:c7 | C     |      | eth0  |
| 10.0.2.3 | ether  | 08:00:27:e5:fd:ed | C     |      | eth0  |
| _gateway | ether  | 52:54:00:12:35:00 | C     |      | eth0  |

```
Entries: 4      Skipped: 0      Found: 4
```

```
root@kali:~# arp -n
```

| Address  | HWtype | HWaddress         | Flags | Mask | Iface |
|----------|--------|-------------------|-------|------|-------|
| 10.0.2.4 | ether  | 08:00:27:ad:87:b3 | C     |      | eth0  |
| 10.0.2.5 | ether  | 08:00:27:27:d6:c7 | C     |      | eth0  |
| 10.0.2.3 | ether  | 08:00:27:e5:fd:ed | C     |      | eth0  |
| 10.0.2.1 | ether  | 52:54:00:12:35:00 | C     |      | eth0  |



# nmap command

**Nmap** is Linux command-line tool for network exploration and security auditing. This tool is generally used by hackers and cybersecurity enthusiasts and even by network and system administrators for **port scanning**. It is used for the following purposes:

- ❑ Real time information of a network
- ❑ Detailed information of all the IPs activated on your network
- ❑ Number of ports open in a network
- ❑ Provide the list of live hosts
- ❑ Port, OS and Host scanning

```
$ sudo apt update  
$ sudo apt-get install nmap
```

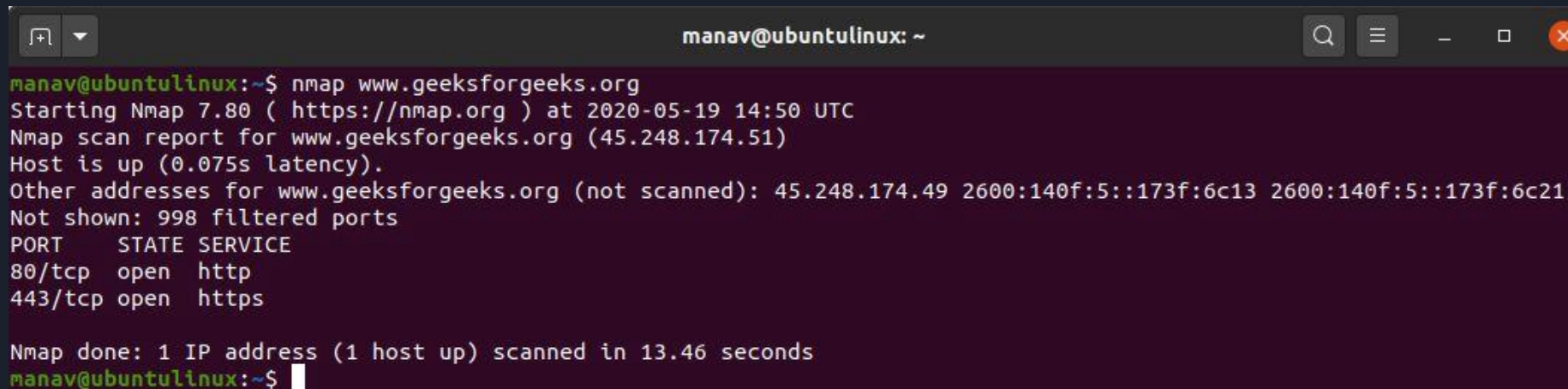
The simplified syntax of the nmap command is as follows:

**nmap [Options] [Target...]**



When invoked as a non-root user that does not have raw packet privileges, nmap runs TCP connect scan. The (-sT) is turned on by default in unprivileged mode.

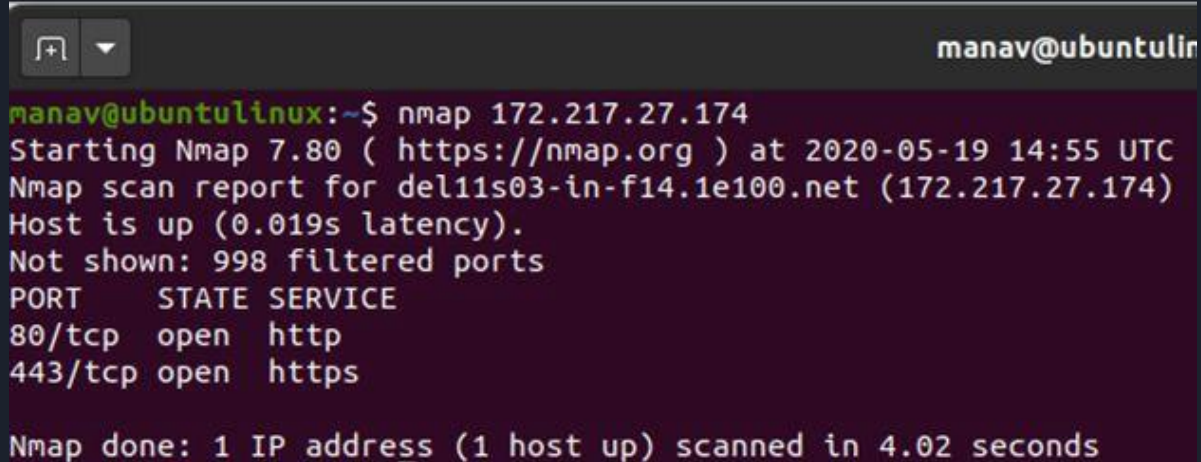
The output will look something like this, including basic information about the scan and a list of open and filtered TCP ports.



```
manav@ubuntulinux: ~  
manav@ubuntulinux:~$ nmap www.geeksforgeeks.org  
Starting Nmap 7.80 ( https://nmap.org ) at 2020-05-19 14:50 UTC  
Nmap scan report for www.geeksforgeeks.org (45.248.174.51)  
Host is up (0.075s latency).  
Other addresses for www.geeksforgeeks.org (not scanned): 45.248.174.49 2600:140f:5::173f:6c13 2600:140f:5::173f:6c21  
Not shown: 998 filtered ports  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 13.46 seconds  
manav@ubuntulinux:~$
```

Source image: <https://www.geeksforgeeks.org/nmap-command-in-linux-with-examples/>

We can also scan using IP Address:

A terminal window with a dark background and light text. The window title bar shows a maximize button, a close button, and the username 'manav@ubuntulin'. The terminal content shows the execution of the 'nmap' command on the IP address '172.217.27.174'. The output indicates that the host is up and lists open ports 80 (http) and 443 (https).

```
manav@ubuntulin:~$ nmap 172.217.27.174
Starting Nmap 7.80 ( https://nmap.org ) at 2020-05-19 14:55 UTC
Nmap scan report for del11s03-in-f14.1e100.net (172.217.27.174)
Host is up (0.019s latency).
Not shown: 998 filtered ports
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 4.02 seconds
```

The nmap command allows scanning a system in various ways. In this we are performing a scan using the hostname as “geeksforgeeks” and IP address “172.217.27.174”, to find all open ports, services, and MAC addresses on the system.

We can also scan using “-v” option, to get more verbose information.



We can scan multiple hosts by writing IP addresses or hostnames with nmap.

We can do this by just writing the multiple IP addresses with a space in between .

```
manav@ubuntulinux: ~  
manav@ubuntulinux:~$ nmap 103.76.228.244 157.240.198.35 172.217.27.174  
Starting Nmap 7.80 ( https://nmap.org ) at 2020-05-19 16:57 UTC  
Nmap scan report for bridgei2p.com (103.76.228.244)  
Host is up (0.062s latency).  
Not shown: 991 filtered ports  
PORT      STATE SERVICE  
22/tcp    open  ssh  
25/tcp    open  smtp  
80/tcp    open  http  
110/tcp   open  pop3  
443/tcp   open  https  
465/tcp   open  smtps  
587/tcp   open  submission  
993/tcp   open  imap  
995/tcp   open  pop3s  
  
Nmap scan report for edge-star-mini-shv-01-del1.facebook.com (157.240.198.35)  
Host is up (0.040s latency).  
Not shown: 998 filtered ports  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap scan report for kix05s07-in-f174.1e100.net (172.217.27.174)  
Host is up (0.041s latency).  
Not shown: 998 filtered ports  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 3 IP addresses (3 hosts up) scanned in 12.96 seconds
```



To scan whole subnet:

```
nmap 103.76.228.*
```

We can scan a whole subnet or IP range with nmap by providing “\*” with it.

It will scan a whole subnet and give the information about those hosts which are Up in the Network.

## -iL option:

If we have a long list of addresses that we need to scan, we can directly import a file through the command line. It will produce a scan for the given IP addresses.

```
manav@ubuntulinux: ~/gfg
manav@ubuntulinux:~/gfg$ nmap -iL input.txt
Starting Nmap 7.80 ( https://nmap.org ) at 2020-05-19 17:52 UTC
Nmap scan report for bridge12p.com (103.76.228.244)
Host is up (0.095s latency).
Not shown: 954 filtered ports, 32 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
110/tcp   open  pop3
143/tcp   open  imap
443/tcp   open  https
465/tcp   open  smtps
587/tcp   open  submission
993/tcp   open  imaps
995/tcp   open  pop3s
2222/tcp  open  EtherNetIP-1
3306/tcp  open  mysql

Nmap done: 1 IP address (1 host up) scanned in 25.60 seconds
```



## OS, Service and Version Detection

Nmap can detect the remote host operating system using TCP/IP stack fingerprinting. To run OS detection, invoke the command with the `-O` option:

```
sudo nmap -O scanme.nmap.org
```

If Nmap can detect the host OS, it will print something like below:

```
Device type: general purpose
```

```
Running: Linux 5.X
```

```
OS CPE: cpe:/o:linux:linux_kernel:5
```

```
OS details: Linux 5.0 - 5.4
```

```
Network Distance: 18 hops
```

```
OS detection performed. Please report any incorrect results at https://nmap.org/subr
```

```
Nmap done: 1 IP address (1 host up) scanned in 26.47 seconds
```



# SSH (Secure Shell)

- ❑ SSH is a protocol for secure remote access to a machine over untrusted networks.
- ❑ It provides secure transmission.
- ❑ Improvement over Telnet as telnet used to send data as plain text and host between sender and receiver can see the traffic.
- ❑ It provides secure command shell (SCP), it allows us to transfer a file securely.
- ❑ Diffie Hellman key exchange algorithm is used to implement SSH
- ❑ It provides strong encryption and server authentication.
- ❑ Server normally listens for connections on port 22.



# SSH Commands

**Basic Syntax:** `ssh [user]@[hostname]`

Example: `ssh username@server.com`

**Logging in with Key-Based Authentication:**

`ssh -i /path/to/private/key username@server.com`

**Common SSH Options:**

- p: Specify a custom SSH port
- l: Specify a custom username
- v: Verbose mode for debugging
- X: Enable X11 forwarding (for GUI applications)

**Close SSH Session:** Simply type 'exit' or press Ctrl + D.



```
(base) shashankrustagi@Shashanks-MacBook-Air ~ % ssh iiitd@192.168.2.233
iiitd@192.168.2.233's password:
Welcome to Ubuntu 18.04.4 LTS (GNU/Linux 4.15.0-76-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/advantage
```

```
System information as of Thu Oct  7 11:46:58 IST 2021
```

```
System load:  0.0           Processes:            98
Usage of /:   18.8% of 15.68GB Users logged in:      0
Memory usage: 27%          IP address for ens32: 192.168.2.233
Swap usage:   3%
```

```
* Canonical Livepatch is available for installation.
- Reduce system reboots and improve kernel security. Activate at:
  https://ubuntu.com/livepatch
```

```
50 updates can be applied immediately.
26 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
```

```
New release '20.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
```

```
Last login: Thu Oct  7 11:24:50 2021 from 192.168.1.99
iiitd@ubuntu:~$
```

# Secure Copy (SCP)

SCP (secure copy) is a command-line utility that allows you to securely copy files and directories between two locations.

With `scp`, you can copy a file or directory:

- From your local system to a remote system.
- From a remote system to your local system.
- Between two remote systems from your local system.

When transferring data with `scp`, both the files and password are encrypted so that anyone snooping on the traffic doesn't get anything sensitive.

In this tutorial, we will show you how to use the `scp` command through practical examples and detailed explanations of the most common `scp` options.



### Command:

```
scp [OPTION] [user@]SRC_HOST:]file1  
[user@]DEST_HOST:]file2
```

`scp` provides a number of options that control every aspect of its behavior. The most widely used options are:

- `-P` - Specifies the remote host ssh port.
- `-p` - Preserves files modification and access times.
- `-q` - Use this option if you want to suppress the progress meter and non-error messages.
- `-C` - This option forces `scp` to compresses the data as it is sent to the destination machine.
- `-r` - This option tells `scp` to copy directories recursively.

The `scp` command relies on `ssh` for data transfer, so it requires an ssh key or password to authenticate on the remote systems.



To copy a file from a local to a remote system run the following command:

Where `file.txt` is the name of the file we want to copy, `remote_username` is the user on the remote server, `10.10.0.2` is the server IP address. The `/remote/directory` is the path to the directory you want to copy the file to. If you don't specify a remote directory, the file will be copied to the remote user home directory.

You will be prompted to enter the user password, and the transfer process will start.

```
remote_username@10.10.0.2's password:
file.txt          100%    0    0.0KB/s    00:00
```

```
(base) shashankrustagi@Shashanks-MacBook-Air Desktop % scp file1.txt iiitd@192.168.2.233:
iiitd@192.168.2.233's password:
file1.txt
```

100% 18 0.1KB/s 00:00

Omitting the filename from the destination location copies the file with the original name. If you want to save the file under a different name, you need to specify the new file name:

```
$ scp file.txt remote_username@10.10.0.2:/remote/directory/newfilename.txt
```

If SSH on the remote host is listening on a port other than the default 22 then you can specify the port using the `-P` argument:

```
(base) shashankrustagi@Shashanks-MacBook-Air Desktop % scp -P 233 file1.txt iiitd@192.168.2.233:
ssh: connect to host 192.168.2.233 port 233: Connection refused
lost connection
(base) shashankrustagi@Shashanks-MacBook-Air Desktop %
```

The command to copy a directory is much like as when copying files. The only difference is that you need to use the `-r` flag for recursive.

```
(base) shashankrustagi@Shashanks-MacBook-Air Desktop % scp -r folder iiitd@192.168.2.233:
iiitd@192.168.2.233's password:
```



THANK YOU :)